Advanced NAT/PAT Investigation

Item	Details
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Tool used	Cisco Packet Tracer

1. Introduction

This report explains how a private company network was connected to an external network using Network Address Translation (NAT) and Port Address Translation (PAT). Private IP addresses cannot be used on the internet, so the company router changes them into public addresses before traffic leaves the network. The router also changes them back when replies come in.

1.1 Task description

Configure a private enterprise network that accesses an external network through NAT/PAT. Implement static NAT for an internal server, dynamic NAT for selected hosts, and PAT for normal users. Use packet captures and routing tables to document every translation.

1.2 Objectives

- Build the network in Cisco Packet Tracer.
- Configure static NAT so the internal server can be reached from outside.
- Configure dynamic NAT so selected hosts use a pool of public addresses.
- Configure PAT so normal users share one public address.
- Prove each translation using packet captures, NAT tables and routing tables.

2. Key Concepts in Simple English

Term	Meaning	Everyday example
Private address	An address used only inside the company (for example 192.168.10.x). It cannot be used on the internet.	A room number inside a building.
Public address	An address that works on the outside network (here 203.0.113.x).	The street address of the building.
NAT	The router swaps a private address for a public one.	A receptionist who forwards mail using the building address.
Static NAT	One private address is always linked to one fixed public address.	A permanent mailbox for one person.
Dynamic NAT	A private host borrows a public address	Borrowing a visitor parking space.

Term	Meaning	Everyday example
	from a pool. It is given back when not in use.	
PAT (overload)	Many private hosts share one public address. Port numbers tell them apart.	One office phone number with many extensions.

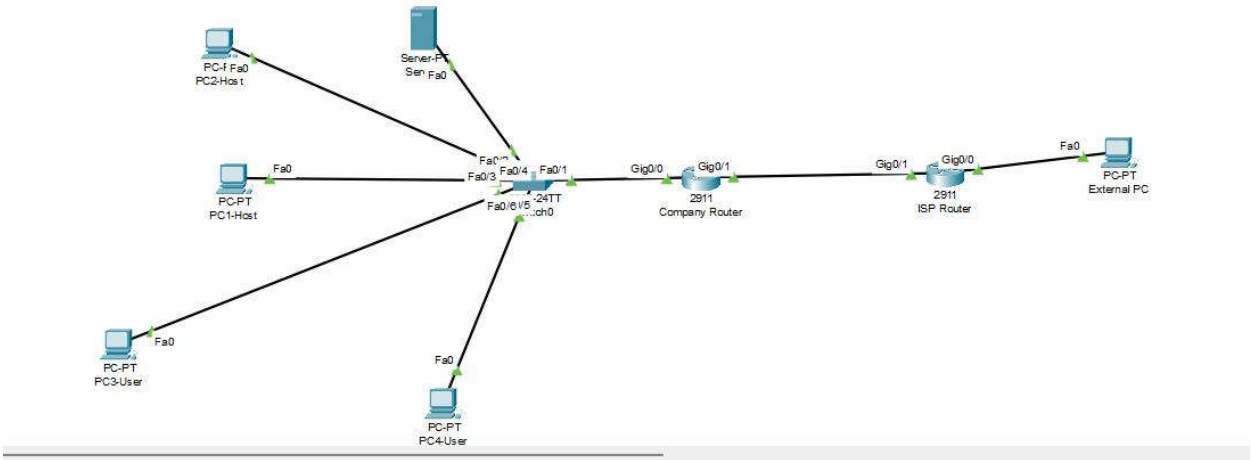
3. Network Design

3.1 Devices used

- R1 (Cisco 2911): the company router. It performs all NAT/PAT.
- R2 (Cisco 2911): the ISP router, which represents the outside world.
- SW1 (Cisco 2960): the internal switch.
- Server0: the internal server.
- PC1 and PC2: selected hosts (dynamic NAT).
- PC3 and PC4: normal users (PAT).
- ExtPC: a computer on the external network.

3.2 Topology

Server0, PC1, PC2, PC3 and PC4 connect to SW1. SW1 connects to R1 Gi0/0 (inside). R1 Gi0/1 (outside) connects to R2 Gi0/1. R2 Gi0/0 connects to ExtPC.



3.3 IP address plan

Device	IP address	Subnet mask	Default gateway
R1 Gi0/0 (inside)	192.168.10.1	255.255.255.0	-
R1 Gi0/1 (outside)	203.0.113.1	255.255.255.0	-
R2 Gi0/1	203.0.113.2	255.255.255.0	-
R2 Gi0/0	198.51.100.1	255.255.255.0	-

Device	IP address	Subnet mask	Default gateway
Server0	192.168.10.10	255.255.255.0	192.168.10.1
PC1 (selected)	192.168.10.11	255.255.255.0	192.168.10.1
PC2 (selected)	192.168.10.12	255.255.255.0	192.168.10.1
PC3 (normal)	192.168.10.100	255.255.255.0	192.168.10.1
PC4 (normal)	192.168.10.101	255.255.255.0	192.168.10.1
ExtPC	198.51.100.10	255.255.255.0	198.51.100.1

3.4 NAT address plan

NAT type	Inside hosts	Public address(es)
Static NAT	Server0 (192.168.10.10)	203.0.113.10 (fixed)
Dynamic NAT	PC1, PC2	Pool: 203.0.113.20 to 203.0.113.25
PAT	PC3, PC4	203.0.113.1 (R1 outside interface) with port numbers

4. How the Task Was Done

Step 1: Build the topology

1. Open Cisco Packet Tracer and start a new file.
2. Add two 2911 routers, one 2960 switch, one server and five PCs.
3. Connect them with copper straight-through cables as shown in section 3.2.
4. Rename the devices (R1, R2 as ISP, SW1, Server0, PC1 to PC4, ExtPC).

Step 2: Set the IP addresses on hosts

For each PC and the server, open Desktop > IP Configuration and enter the IP address, mask and default gateway from section 3.3. The gateway is always R1 Gi0/0 (192.168.10.1) for inside devices, so that traffic for outside networks is sent to the router.

Step 3: Configure the ISP router (R2)

R2 only needs IP addresses on its two interfaces. It does not know about the private 192.168.10.0 network. This is intentional: the outside world must only see public addresses.

```
enable
configure terminal
hostname ISP
interface gigabitEthernet0/1
 ip address 203.0.113.2 255.255.255.0
 no shutdown
interface gigabitEthernet0/0
 ip address 198.51.100.1 255.255.255.0
 no shutdown
end
```

```
write memory
```

Step 4: Configure interfaces on the company router (R1)

Each R1 interface is labelled as "inside" (private side) or "outside" (public side). NAT only works on interfaces that have these labels.

```
enable
configure terminal
hostname R1
interface gigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown
interface gigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.0
 ip nat outside
 no shutdown
ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

The last line is a default route. It tells R1 to send any traffic for unknown networks to R2.

Step 5: Configure static NAT

This line permanently links the server's private address to a public address. Outside users can now reach the server at 203.0.113.10.

```
ip nat inside source static 192.168.10.10 203.0.113.10
```

Step 6: Configure dynamic NAT

First a pool of public addresses is created. Then an access list (ACL 1) names the selected hosts, PC1 and PC2. Finally the two are joined: hosts in ACL 1 may use the pool.

```
ip nat pool SELECTED 203.0.113.20 203.0.113.25 netmask 255.255.255.0
access-list 1 permit host 192.168.10.11
access-list 1 permit host 192.168.10.12
ip nat inside source list 1 pool SELECTED
```

Step 7: Configure PAT

ACL 2 matches the normal users, PC3 (.100) and PC4 (.101). The word "overload" turns on PAT, so they all share the outside address of R1.

```
access-list 2 permit 192.168.10.100 0.0.0.1
ip nat inside source list 2 interface gigabitEthernet0/1 overload
end
write memory
```

Step 8: Save the project

In Packet Tracer choose File > Save As and save as NAT_PAT_Investigation.pkt.

5. Testing and Evidence

Before each test, run "clear ip nat translation *" on R1 so the table starts empty.

5.1 Connectivity check

From PC1, PC3 and Server0 ping 198.51.100.10 (ExtPC). Replies prove that the network path works.

5.2 Dynamic NAT test (PC1 and PC2)

5. On PC1 open Command Prompt and run: ping 198.51.100.10
6. On R1 run: show ip nat translations

Expected result: PC1 (192.168.10.11) appears as an address from 203.0.113.20 to 203.0.113.25. Each selected host gets its own public address.

```
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.20:1    192.168.10.11:1    198.51.100.10:1    198.51.100.10:1
```

5.3 PAT test (PC3 and PC4)

Ping 198.51.100.10 from PC3 and PC4, then run show ip nat translations on R1.

Expected result: both PCs show the same public address, 203.0.113.1, with different port numbers.

```
icmp 203.0.113.1:5   192.168.10.100:5  198.51.100.10:5   198.51.100.10:5
icmp 203.0.113.1:6   192.168.10.101:6  198.51.100.10:6   198.51.100.10:6
```

5.4 Static NAT test (server)

From ExtPC ping 203.0.113.10. The server replies. The permanent entry stays in the NAT table even when no traffic is running.

```
--- 203.0.113.10    192.168.10.10    ---    ---
```

Pinging 192.168.10.10 directly from ExtPC fails, because private addresses are not known on the outside network.

5.5 Statistics

```
show ip nat statistics
```

This shows how many translations are active, how many hits and misses occurred, and which interfaces are inside and outside.

. Packet Captures (Simulation Mode)

Simulation mode lets us watch a packet move step by step and see how the addresses change.

7. Click the Simulation tab at the bottom right of Packet Tracer.
8. Click Edit Filters and select only ICMP.
9. Send a ping from PC1 to ExtPC and press Capture/Forward.
10. Click the packet at SW1 and R1, and open Inbound PDU Details and Outbound PDU Details.

Traffic	Before R1 (inside)	After R1 (outside)	Translation
PC1 to ExtPC	Src 192.168.10.11 Dst 198.51.100.10	Src 203.0.113.20 Dst 198.51.100.10	Dynamic NAT
PC3 to ExtPC	Src 192.168.10.100 Dst 198.51.100.10	Src 203.0.113.1 Dst 198.51.100.10	PAT
ExtPC to server	Src 198.51.100.10 Dst 203.0.113.10 (arrives at R1)	Src 198.51.100.10 Dst 192.168.10.10	Static NAT
Server to ExtPC	Src 192.168.10.10 Dst 198.51.100.10	Src 203.0.113.10 Dst 198.51.100.10	Static NAT

---

7. Routing Tables

Run show ip route on both routers.

7.1 R1 (company router)

```
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
C    203.0.113.0/24  is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 203.0.113.2
```

R1 knows its two directly connected networks (C) and has a static default route (S*) toward the ISP.

[Insert screenshot: show ip route on R1]

7.2 R2 (ISP router)

```
C    198.51.100.0/24 is directly connected, GigabitEthernet0/0
C    203.0.113.0/24  is directly connected, GigabitEthernet0/1
```

R2 has no route to 192.168.10.0/24. This proves the private network is hidden from the outside. Replies still work because they are sent to the translated public addresses in 203.0.113.0/24.

[Insert screenshot: show ip route on R2]

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8. Results Summary

Inside host	NAT type	Seen outside as	Notes
Server0 192.168.10.10	Static NAT	203.0.113.10	Permanent, reachable from outside
PC1, PC2	Dynamic NAT	203.0.113.20-25	One public address each, from the pool

Inside host	NAT type	Seen outside as	Notes
PC3, PC4	PAT	203.0.113.1 + port	Shared address, different ports

9. Troubleshooting Notes

- Ping fails: check each PC's IP address and gateway, and run show ip interface brief to confirm the interfaces are up.
- No translations appear: check that ip nat inside and ip nat outside are on the correct interfaces.
- Only some PCs are translated: check that the ACL matches their addresses (show access-lists).
- Pool runs out: dynamic NAT only has 6 addresses. Use clear ip nat translation * to free them.
- Old entries confuse the test: clear the NAT table before every test.

10. Conclusion

The network was built with a company router (R1) that translates private addresses into public ones. Static NAT gave the server a fixed public address. Dynamic NAT gave selected hosts addresses from a pool. PAT let normal users share one address using port numbers. The NAT tables, packet captures and routing tables confirm that each translation works as designed.